Designed to Protect & Enhance Plastic Products

100% Biodegradable Plastic Technology

d2p
www.d2p.net

d2w®
www.d2w.net

Symphony is a world leader in the development and marketing of additives and masterbatches to make plastic products better

Making Plastic Smarter
Symphony environmental
India

Distributors:
Plasto Films

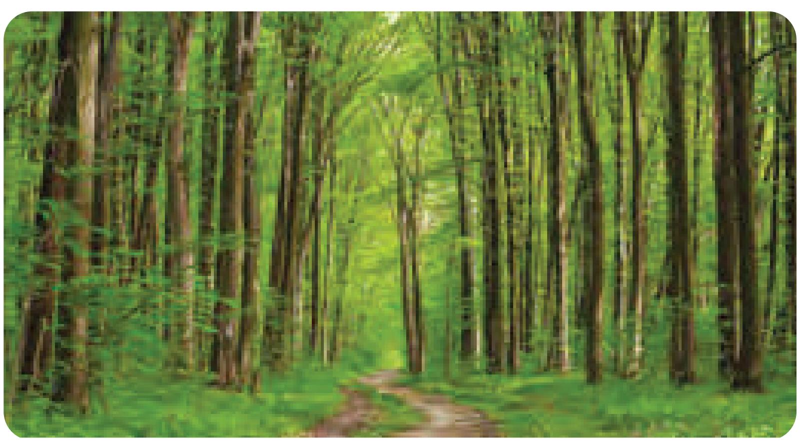
100% Biodegradable Plastic technology

A scientifically proven technology* and environmentally responsible solution for your plastic product, film or packaging needs.

A masterbatch which turns ordinary plastic at the end of its useful life, in the presence of oxygen, into a material with a different molecular structure. At the end of the process, it is no longer a plastic, and instead has changed to a material which is biodegradable (by bacteria and fungi) in the open environment.

Stages of biodegradation with $d_2w^®$ technology:

1. $d_2w^®$ masterbatch is added at the manufacturing stage.
2. Film containing $d_2w^®$ is extruded and then converted into bags or packaging.
3. The product behaves like conventional products during its intended service life.
4. After its service life, the bag or packaging may end up in the open environment.
5. The $d_2w^®$ then takes effect and the product begins to degrade in the presence of oxygen.
6. The product will degrade and biodegrade in a continuous, and irreversible process, leaving nothing but carbon dioxide, water and biomass.



Helping to protect the environment from plastic litter

Disclaimer: The information provided is general information. For specific applications, please consult our Technical Department. The buyer is responsible for advertising and labelling of products made with $d_2w^®$, and for compliance with all applicable laws in the place where such products are to be supplied, advertised, or used.

Added Value with $d_2w^®$

- Requires only 1% inclusion rate.
- Works with virgin and recycled plastic.
- Works with PE & PP.
- No change to the manufacturing process.
- Does not lose any of its original properties duringits useful life.
- Our customers receive full support from Symphony's Technical and Marketing teams.

*TESTED TO THE FOLLOWING INTERNATIONAL STANDARDS

✓ American ASTM D6954
✓ British Standard 8472
✓ British PAS 9017: 2020
✓ French Accord T51-808
✓ Saudi Standard SASO 2879
✓ UAE Standard 5009:2009
✓ EN ISO 14855 Biodegradable in Compost

Approved by

✓ SASO (Saudi Arabia)
✓ ESMA (EMIRATES UAE)

***tested according to ASTM D6954**

How does $d_2w^{®}$ Work?

PE and PP polymers are composed of hydrocarbons, comprising of small units bonded into long chains. These chains of high molecular weight which gives rise to plastics' useful qualities of low mass, strength, and imperviousness.

Symphony's $d_2w^{®}$ acts on the carbon bonds, shortening the chains, meaning the material effectively ceases to be plastic and becomes a food source for microbes.

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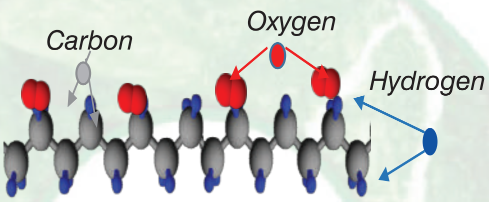
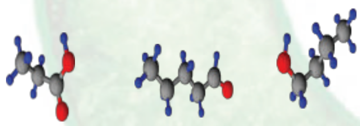
This process of biodegradability is initiated in ambient conditions when exposed to oxygen in the air and accelerated by exposure to UV light and/or increased temperature.

STAGE 1	STAGE 2	STAGE 3
Pro-degradant: metal stearate, thermo and photo oxidant	Oxidation: oxygen/ UV/heat Short chain hydrophilic	CO2+H2O+Biomass Biodegradation



The catalytic effect of the $d_2w^{®}$ additive quickly breaks the long chains and generates free radicals. During the abiotic phase the polymers undergo free radical oxidationreactions which result in short chained molecules.



The free radicals combined with the available oxygen atoms create hydro-peroxides. These shorter chained molecules are now hydrophilic.

The resulting short-chain hydroperoxides are now available for consumption by micro-organisms. The biodegradation process creates harmless residues as CO_2, H_2O and Biomass.

Open Exposure to the Environment

High Mw Polymer with $d_2w^{®}$
~100,000 – 200,000 Da

Failure & Fragmentation
~15,000 Da

Biodegradable Oligomers
< 5,000 – 1,500 Da

Different blends of $d_2w^{®}$ are engineered depending on the end customer requirements, with some highly aggressive blends causing fragmentation in only a few weeks and rapid molecular weight reduction and bio-assimilation in the open environment.

Safe for Recycling

Recycling of 100%-biodegradable Plastics

- 100% Biodegradable plastics are prenominal conventional plastics, with the small addition of prodegradant formulation (<0.2% w/w) and so can be recycled along with conventional recycling streams.
- Repeated testing demonstrates the performance of products made with varying levels of Recyclate, up to 100%

https://www.symphonyenvironmental.com/resource/new-tckt-rport-confirms-oxo-biodegradable-plastic-can-be- recycled-with-ordinary-plasti/

d_2p® at a glance

The d2p® suite of technologies provide cost-effective protection toproducts and surfaces in applications as diverse as food packaging and processing to flame retardants and pest control.

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Antibacterial
An antibacterial masterbatch successfully tested against many common organisms including dangerous bacteria such as MRSA, E.coli, Salmonella, Listeria, Pseudomonas and Aspergillus Niger.

Natural
Antibacterial suitable for use in food and non-food applications. In compliance with FDA Food and Drug Administration, USA and EFSA (European Food Safety Authority) requirements.

Antimicrobial
A family of additives offering cost-effective protection of plastic products fnst contamination, staining, discolouration and odour caused by bacteria and fungi.

Anti-Fouling
Anti-fouling paint is a specialised coating applied to the hull of a ship or boat to reduce the growth of aquatic organisms.

Insecticide Technology
Insecticidal plastic masterbatches used to control pests. Typically used in mosquito nets, agriculture, horticulture, forestry and home applications.

Oxygen Scavenger
d2p®OS is a powerful inorganic chemicalcompound, produced from a natural ore and manufactured to a high purity.

Flame Retardant
Flame retardants decrease the ignitability of materials and inhibit the combustion process – limiting the amount of heat released.

Vapour Corrosion Inhibitor
d_2p® VCI additives are a range of productsto be used in protection of surfaces against the corrosion and oxidation of ferrous and non-ferrous metals.

Ethylene Adsorber
Highly active adsorbent masterbatch for the removal of undesirable odours, volatile organic compounds (VOC) and water vapour from plastic packaging to reduce spoilage of fruit and vegetables.

Odour Adsorber
Inorganic masterbatches and additives designed to inhibit the development of odours in plastic products and reduce spoilage of fruit and vegetables.

Release Agent
A modern synthetic product produced from one of the most common of the earth's elements. It plays a key role in the improvement of the flow and processing of resins as well as enhanc-ing the slip and lubricity of plastic products.

Rodent Repellent
Rodents can cause dangerous damage to plastic products such as cable insulation, warehouse pallets, non-food packaging and boxes etc. Symphony has developed additive masterbatches with products thatrepel these pests.

Purge Technology
A Mechnical and Chemical Cleaning Solution for the Plastic Processing Industry.

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RECYCLED
Paper made from recycled material
FSC® C020119

bsi. ISO 9001 Quality Management | ISO 14001 Environmental Management
FM37939 EMS 567450

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